

LESSON 6-5 • SESSION 1 • PRACTICE • FORM B

Writing Algebraic Expressions

Learning target – I can write an algebraic expression that uses a variable for an unknown amount. · 6.EE.A.2a

REMEMBER

- A **variable** is a letter for an unknown amount. $5n$ means $5 \times n$.
- In $5n + 12$, 5 is the **coefficient** and 12 is the **constant**.
- An amount that **repeats** goes with the variable. "4 less than x " is $x - 4$.

WORKED EXAMPLE

A taxi charges \$3 to start plus \$2 for each mile. Write an expression for the fare for m miles.

Step 1 Let m = the number of miles.

Step 2 \$2 repeats each mile: $2m$.

Step 3 The \$3 start fee happens once. Join: $2m + 3$.

Answer $2m + 3$ dollars

WORDS **variable** (*variable*) · **coefficient** (*coeficiente*) · **constant** (*constante*) · **algebraic expression** (*expresión algebraica*)

PRACTICE

show your work · label units

- Write an algebraic expression for each phrase.
 - 9 more than a number p _____
 - 8 times a number z _____
 - 10 less than r _____
 - b divided by 6 _____
 - three times s , plus 4 _____
 - 2 less than the product of 5 and j _____
- Art Club.** An art club buys one \$9 paint kit for each member. It also pays one \$16 shipping charge. Let m stand for the number of members.
 - Write the paint kit cost as a term. Name its coefficient. _____
 - What is the constant? Why has it no variable? _____
 - Write one expression for the total cost. _____
- Concert Tickets.** Each ticket costs \$11. There is one \$20 service fee for the whole order. Write an expression for the cost of t tickets.
Hint The \$11 repeats; the \$20 happens once.

 Show your work.
 Expression _____
- Car Wash.** A team charges \$6 for each car it washes. It pays one \$35 fee for supplies. Write an expression for the money left after c cars.
Hint Money collected minus the fee.

 Show your work.
 Expression _____
- Phone Plan.** A phone plan costs \$12 each month plus a one-time \$25 activation charge. Let m be the number of months. Write an expression for the cost.
Hint The monthly charge goes with the variable.

 Show your work.
 Expression _____

6. **Bike Rental.** A bike rental costs \$4 for each hour. Tariq wrote $4 + h$ for the cost of h hours. Justify or correct his expression.

Hint Does “for each hour” mean add or multiply?

Show your work.

Expression _____

7. **Two Items.** A stand sells juice for \$2 a cup and muffins for \$3 each. Let j be the cups sold and f be the muffins sold. Write one expression for the money collected.

Hint Write a term for each item, then add.

Show your work.

Expression _____

8. **Compare Gyms.** Gym A charges a \$30 joining fee plus \$20 each month.

- a. Write an expression for Gym A’s cost after m months. _____
- b. Gym B charges no joining fee but \$24 each month. Write its expression. _____
- c. Which gym costs less for 5 months? Show how you know. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Omar’s Work.** Books cost \$9 each and shipping is \$5 one time. Omar wrote $9 + 5k$ for k books. Explain his mistake. Then write the correct expression.

TEST PRACTICE

choose the best answer

10. “5 times a number x ” is written –

A $5 + x$ **B** $x - 5$ **C** $5x$ **D** $x \div 5$

11. “4 less than a number y ” is written –

A $4 - y$ **B** $y - 4$ **C** $4y$ **D** $y + 4$

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

12m + 25 T	4h E	4 + h Q	25m + 12 X	35 – 6c Z	6c – 35 T	11t + 20 E	9m + 16 L	20m + 30 R
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#2c	#3	#4	#5	#6	#8a
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____